

Proof of Concept

EcoRoute

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EcoRoute

AI-Powered Waste Management

[City Council Admin](#)[Fleet Driver](#)

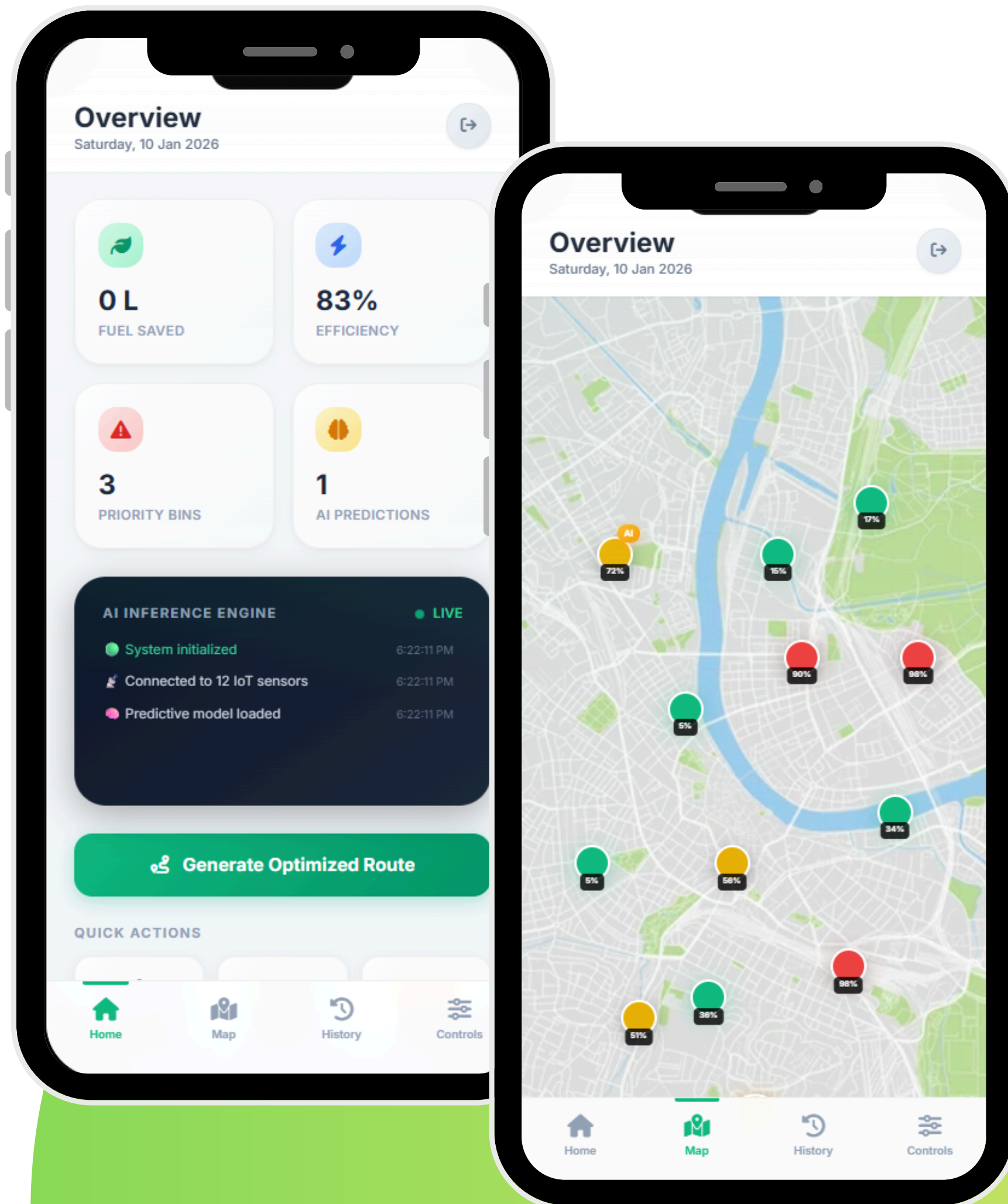
City Council Admin

1. Home Tab

- Displays key performance metrics showing fuel saved, efficiency percentage and priority bins.
- Features an AI inference engine console that displays live sensor data and system logs in real-time.

2. Map Tab

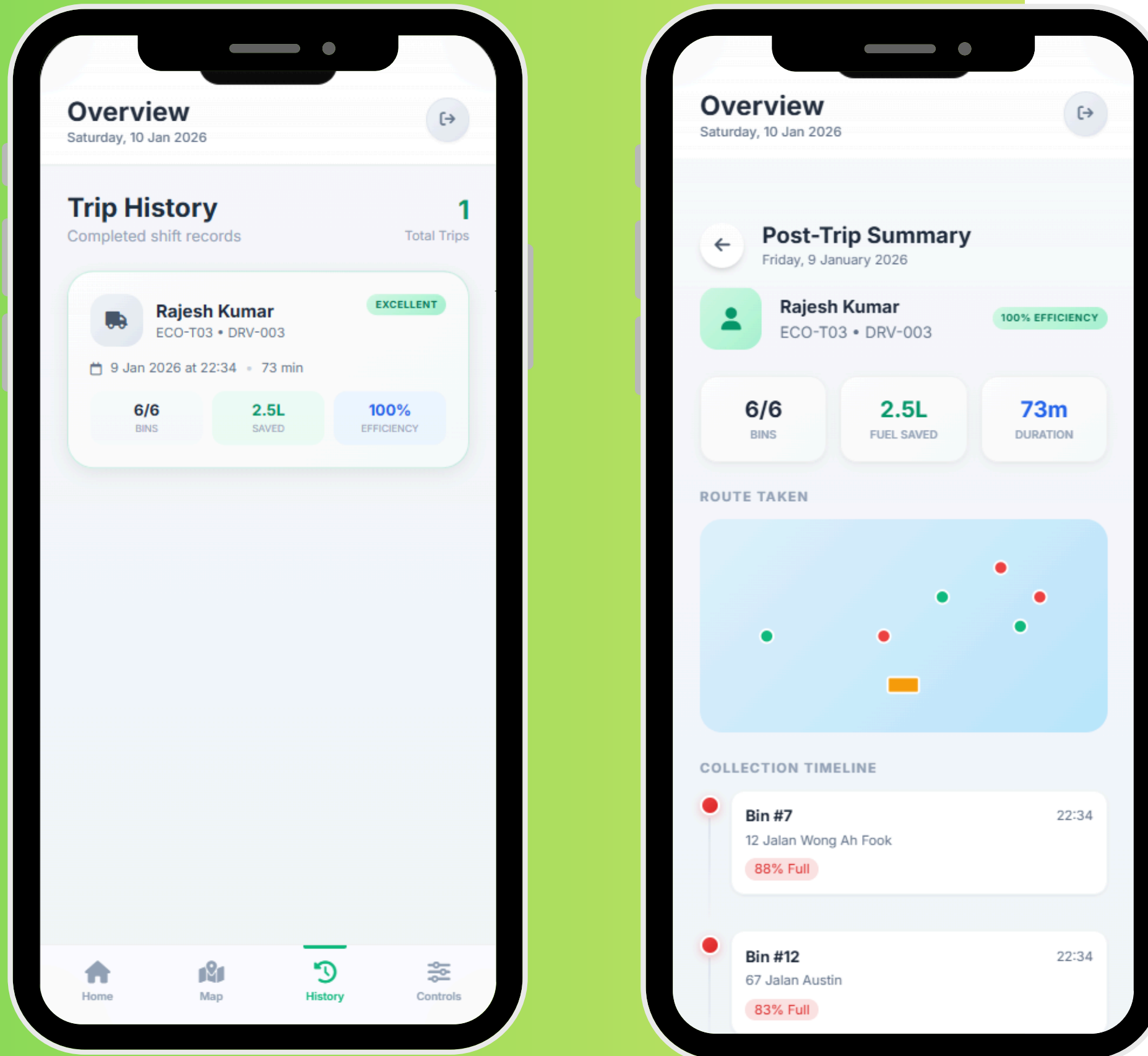
- Showing all waste bin locations across the city.
- Displays color-coded bin markers (green for low fill, yellow for medium, red for critical)

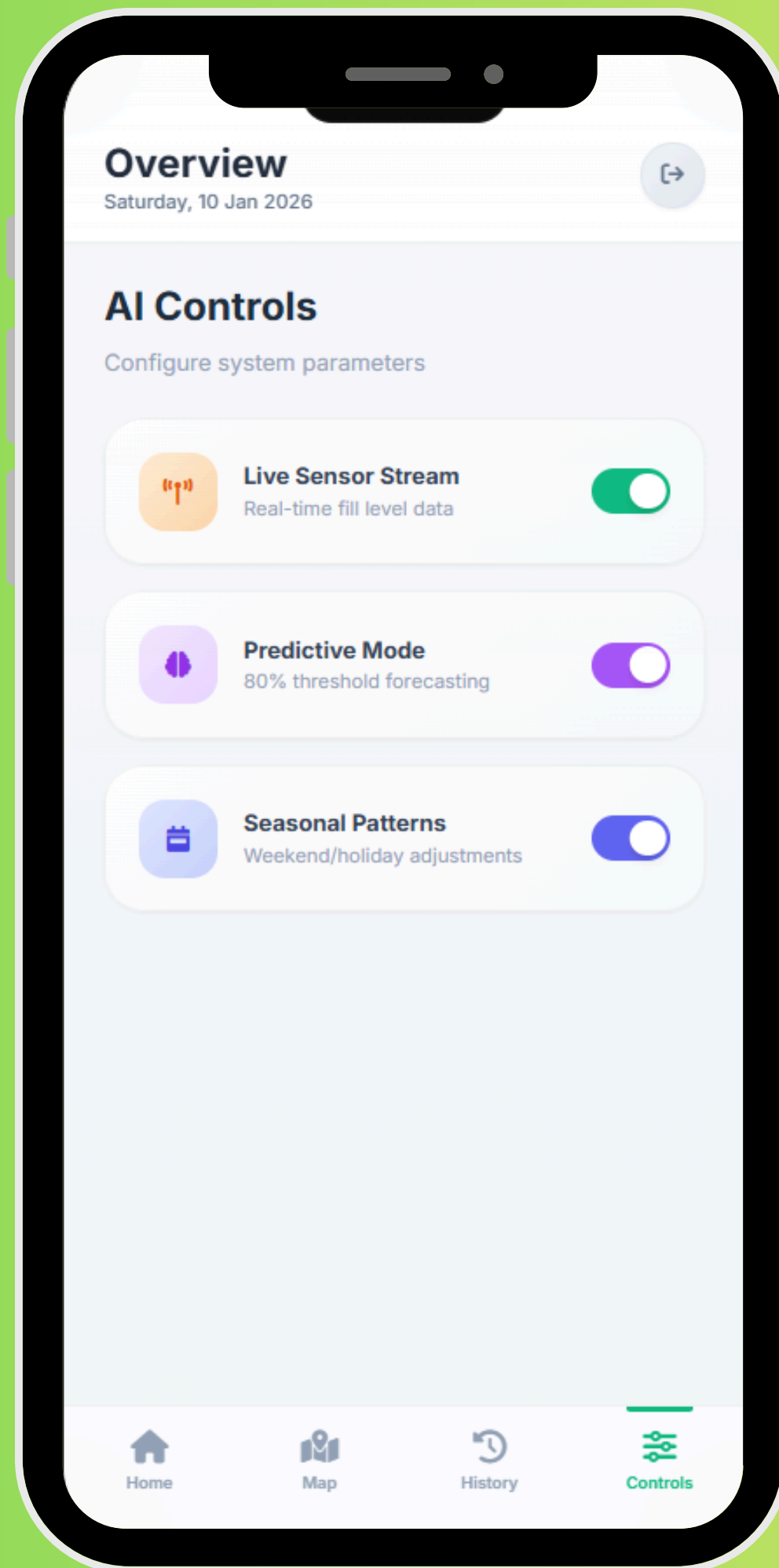


City Council Admin

3. History Tab

- Completed collection shifts displaying trip cards with date, driver name, vehicle ID, bins collected, and efficiency rating.
- Each trip card is clickable to reveal a detailed post-trip summary showing route map, collection timeline, metrics (bins/fuel/duration), and operational override logs for audit purposes. Shows total trip count at the top.





City Council Admin

4. Controls Tab

- **Live Sensor Stream:** Activates real-time fill level monitoring from IoT sensors
- **Predictive Mode:** Enables AI forecasting to predict which bins will reach 80% capacity
- **Seasonal Patterns:** Adjusts predictions based on weekend/holiday waste generation patterns

Fleet Driver

1. Pre-Shift State

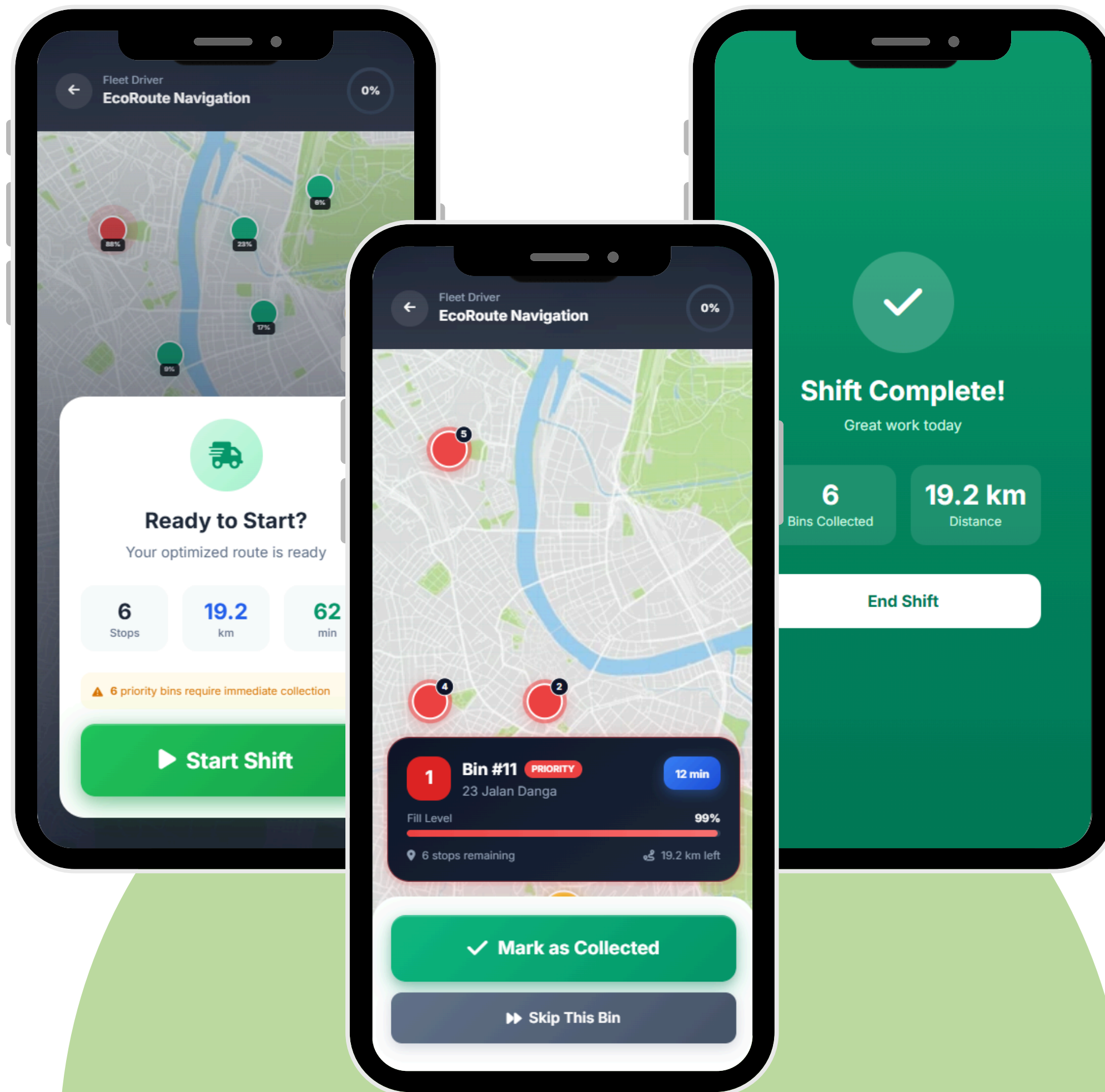
- Displayed before the driver begins their route.
- Route summary statistics : total number of stops, estimated distance, estimated time

2. Active Shift State

- Shows the active bin details, address, fill level percentage with visual progress bar
- Mark as Collected" to confirm pickup and advance to the next stop

3. Completed State

- Shows the total bins collected and total distance traveled during the shift.



Thank You

